

Total No. of Questions : 8]

SEAT No. :

PE-2154

[Total No. of Pages : 2

[6584]-53

B.E. (Civil Engineering)
DAMS AND HYDRAULICS STRUCTURES
(2019 Pattern) (Semester - VIII) (401011)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 OR Q.2, Q.3 OR Q.4, Q.5 OR Q.6, Q.7. OR Q.8.
- 2) Neat sketches/diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks for the sub-questions.
- 4) Assume suitable data if necessary and state them in your answer clearly.
- 5) Use non-programmable pocket size electronic calculator is allowed.

Q1) a) What are spillways and what is their necessity? [5]

b) Draw a labeled sketch of ogee spillway showing all components [5]

c) Explain 'Stilling Basin Types' according IS 4997-1968 [7]

OR

Q2) a) Explain importance of 'Energy Dissipators'. [5]

b) Explain in details 'Vertical Lift Gate'. [5]

c) Design an ogee spillway for concrete gravity dam, for the following data. [7]

i) Average river bed level = 100 m

ii) Slope of D/S = 0.7 H: 1V, u/s face is vertical

iii) Spillway crest RL = 204 m

iv) Design discharge = 8000 m³/s

v) Spillway length is 6 spans with a clear length of 10 m each.

Pier thickness = 2.5 m.

Q3) a) Explain function of pitching on U/S face of earthen dam & turfing. [5]

b) Explain in details homogenous embankment type of dam. [5]

c) Describe the method of plotting phreatic line for an earth dam with & without horizontal filter at the downstream side. [8]

P.T.O.

OR

- Q4)** a) Explain significance of drainage blanket in earthen dam [5]
b) Explain 'Seepage Failure' in earthen dam [5]
c) Determine the factor of safety of downstream slope of (homogeneous section) an earth dam drawn to a scale of 1cm = 4 m, for the following data : [8]

Area of N-rectangle = 24 cm²

Area of T-rectangle = 12 cm²

Area of U-rectangle = 8 cm²

Length of slip circle arc = 20 cm

angle of internal friction = 25°

Cohesion $c = 19.5 \text{ kN/m}^2$

specific weight of soil = 19 kN/m³

- Q5)** a) Explain types of canals based on alignment. [5]
b) Explain in details limitations of Kennedy's theory. [5]
c) Design an irrigation channel by Kennedy's theory to carry a discharge of 36 cumecs. Take rugosity coefficient $N = 0.0225$ and critical velocity ratio $m = 1.05$. The channel has a bed slope of 1 in 5000 and side slope of 0.5 H:1 V. Take initial trial depth as 2m. [8]

OR

- Q6)** a) Explain various losses in irrigation of canals [5]
b) Explain the concept of canal fall and describe ogee fall in details [5]
c) Design a regime channel of trapezoidal section for carrying water at the rate 12 cumecs having side slopes 1 H: 2 V, if Lacey's silt factor is 0.90. [8]

- Q7)** a) Explain in details "Super Passage". [5]
b) Explain in details limitation of Bligh's Theory. [5]
c) Draw layout plan of a typical diversion head works and label all its part. Write functions of each part. [7]

OR

- Q8)** a) Explain in details "Level Crossing". [5]
b) Explain selection criteria for 'Diversion Head Work'. [5]
c) Explain 'Creep Theory by Bligh' for the hydraulic design of weir on a permeable foundation. [7]

